Quantum Solid State Chip — Fabrication Protocol

Same chip architecture, middle block = 4H-SiC wafer with V_Si ($2|D| \approx 70$ MHz, $T_2 \approx 40$ µs @ 300 K) · on-chip CIS/ZnS QD pump (730–760 nm, no big diode) + 850 nm filter

QSSC v2.0.0
SiC variant v1.0
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Quantum Solid State Chip (Room Temperature) — Fabrication Protocol · SiC variant

Project: QSSC (Quantum Solid State Chip, room-temperature demonstrator) **Document type:** Laboratory fabrication protocol (standalone variant) **Version:** 1.0 **Variant:** Silicon-carbide silicon-vacancy (V_Si, V2 line in 4H-SiC) **Scope:** Same chip architecture as the hBN base protocol — (1) an electrically-pumped quantum-dot emitter array, (2) SU-8/PDMS polymer waveguides, (3) an on-chip 2D-material (graphene/MoS_2) photodetector array with FPGA readout and software pulse-capture calibration — but the **middle spin-host block is a commercial 4H-SiC wafer hosting V_Si centres** (S=3/2, $2|D| \approx 70$ MHz, ZPL ≈ 916 nm, $T_2 \approx 40$ µs at 300 K). Unlike the diamond variant (which reuses the green CQD emitter unchanged), the SiC variant requires **three real changes** to the optical chain: (i) the on-chip emitter must shift from green (532 nm) to **red/NIR (~730–760 nm)** to pump V_Si — done with an **on-chip $CuInS_2$/ZnS (CIS) colloidal quantum-dot LED, not a big diode**; (ii) the optical filter must shift from a 650 nm long-pass to an **850 nm long-pass** (the 650 would pass the 730–760 pump); (iii) the detector leans on **graphene** (the 916 nm PL is in the IR where MoS_2's monolayer gain drops).

Note. The SiC variant is the **most reworked** of the three: the host is easier (commercial wafer, defects as-grown + irradiation, ~20× better coherence than hBN), but the on-chip emitter is harder. **Carbon CQDs (citrato+urea) cannot reach 730–760 nm** — their ceiling is green ~540 nm. Pumping V_Si on-chip requires **semiconductor colloidal QDs ($CuInS_2$/ZnS, "CIS")**, a hot-injection colloidal synthesis with indium/copper/sulfur precursors under inert atmosphere. This is a real step up in chemistry from boiling citric acid, but it keeps the **on-chip, no-big-diode** constraint the user requires. The divacancy (V_SiV_C, D ≈ 1.3 GHz, T_2 ~ ms with DD, ZPL ~1.1 µm) is noted as an optional upgrade but is **not** the primary protocol here (the QSSC simulator profiles V_Si, 70 MHz).

On-chip emitter (no big diodes). This variant uses an **on-chip CIS/ZnS quantum-dot LED** as the ~730–760 nm pump. No external laser diode. A commercial 730/808 nm diode would be simpler but is explicitly excluded by the design constraint.

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1. Abstract

This document specifies a complete, equipment-minimal fabrication protocol for a room-temperature solid-state quantum photonic chip (QSSC) built on a **4H-SiC wafer hosting V_Si centres**. The chip combines three functional blocks on a single 25 × 25 mm² carrier: (i) a **red/NIR (~730–760 nm)** $CuInS_2$/ZnS colloidal quantum-dot (CIS-QD) emitter array in a polymeric QLED stack (ITO/PEDOT:PSS/poly-TPD/CIS-ZnS/ZnO/Ag) that pumps the V_Si (optical excitation band below its 916 nm ZPL) — no external pump laser; (ii) a **commercial 4H-SiC wafer** with V_Si (V2) centres ($2|D| \approx 70$ MHz, ZPL 916 nm, $T_2 \approx 40$ µs at 300 K), defects formed by as-grown + electron/proton irradiation + anneal; and (iii) SU-8/PDMS waveguides routing the 916 nm V_Si PL through an **850 nm long-pass filter** to an **on-chip graphene/MoS_2 photodetector array** read directly by the FPGA, with software pulse-capture calibration (no camera, no alignment jig). A Peltier holds the wafer at 288–300 K. Equipment footprint ~USD 3,500–4,500 (the CIS hot-injection setup adds a Schlenk line / Ar glovebag over the carbon-CQD autoclave).

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2. Introduction & Scientific Basis

2.1 Why SiC (V_Si) for the same chip

The silicon vacancy V_Si (V2 line) in 4H-SiC is a **spin-3/2** (S=3/2) defect with zero-field splitting **$2|D| \approx 70$ MHz**, ZPL ≈ 916 nm, optically addressable and coherently controlled at room temperature. Its ensemble Hahn-echo coherence is **$T_2 \approx 40$ µs at 300 K** (natural abundance), extending to ~100 µs in

isotopically purified 4H-SiC and further with dynamical decoupling — roughly **20× better than hBN V_B^-** (2 µs). Critically, the host is a **commercial semiconductor wafer** (4H-SiC power-electronics wafers are mass-produced and cheap), and the defects are created by **standard irradiation + anneal** — no liquid exfoliation, no 2D transfer. This is the **easiest host to fabricate** in the QSSC family.

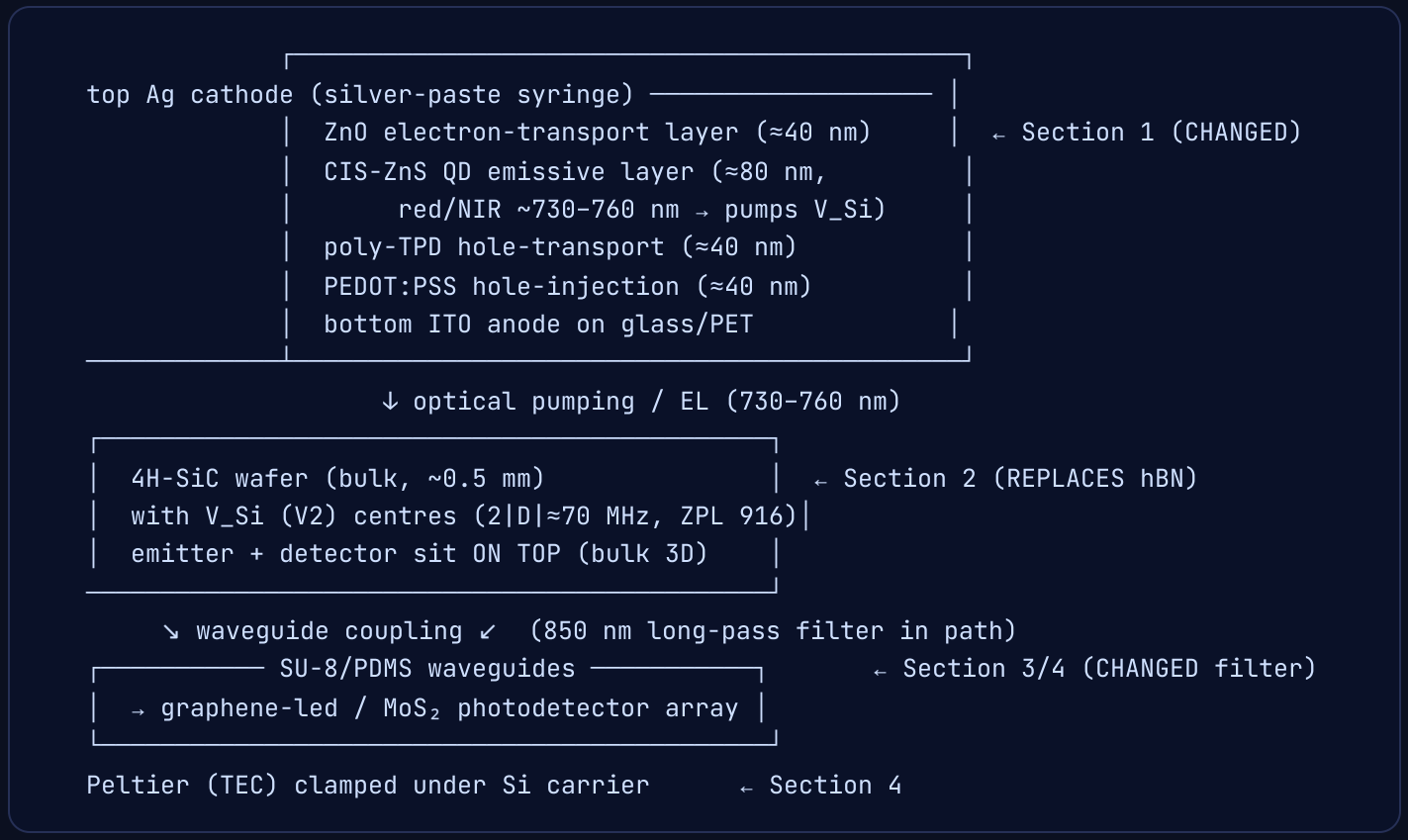
2.2 Why the emitter must change (and why not carbon CQDs)

V_Si is optically excited **below its 916 nm ZPL** — the standard pump is a **730 nm** red laser (or 808 nm, or white light). The hBN chip's green (532 nm) CQD emitter is **sub-optimal** for V_Si (532 nm is sub-bandgap for SiC and only weakly excites the defect). **Carbon quantum dots (citric-acid+urea route) cannot reach 730–760 nm** — their realistic ceiling is green ~540 nm; "red carbon dots" are an active research area with low, unreproducible quantum yield. To keep the **on-chip, no-big-diode** constraint, the emitter is switched to **$CuInS_2$/ZnS (CIS) colloidal QDs**, a Cd/Pb-free semiconductor-dot system that emits tunably across **550–800+ nm** and is the established "green" (heavy-metal-free) NIR-QLED material. CIS QDs at ~730–760 nm pump V_Si directly.

2.3 Why the filter and detector must change

V_Si PL is at **916 nm** (NIR). The hBN chip's **650 nm red long-pass would pass the 730–760 nm pump** (730–760 > 650) and fail to separate pump from signal. So the filter shifts to an **850 nm long-pass** (passes 916 nm PL, blocks 730–760 nm pump). On the detector side, the 916 nm PL is in the IR: monolayer MoS_2 (bandgap ~690 nm) does **not** absorb 916 nm, so the MoS_2 photogating gain is lost and the readout relies on **broadband graphene** (or multilayer MoS_2, bandgap ~1.2 eV ≈ 1030 nm, which does absorb 916). Graphene absorbs 916 nm comfortably — the detector works, but the channel is graphene-led here, unlike the hBN/diamond cases where MoS_2 contributes gain.

2.4 Architecture overview



3. Materials & Bill of Materials (BOM) — deltas from the hBN base

3.1 Added / substituted reagents

ITEM	SPEC / SOURCE	QTY	APPROX. COST
4H-SiC wafer (semi-insulating or N-doped, 2" or 5×5 mm sample, double-side polished)	WolSpeed / II-VI / Alibaba 4H-SiC	1 sample	$30–$120
(Optional, if wafer has no V_Si) electron/proton irradiation service	—	1 run	$0 (university) – $200
Copper(I) iodide (CuI, 99.999 %)	Sigma 205568	5 g	$40
Indium(III) acetate ($In(OAc)_3$, 99.99 %)	Sigma 510270	5 g	$60
1-Dodecanethiol (≥98 %) — sulfur source / ligand	Sigma 471364	100 mL	$30
Oleylamine (70 %, technical) — high-boiling ligand/solvent	Sigma O7805	100 mL	$30
Oleic acid (90 %) — co-ligand	Sigma O1008	100 mL	$20
1-Octadecene (90 %) — non-coordinating solvent	Sigma O806	100 mL	$25
Zinc stearate (or zinc acetate + sulfur) — ZnS shell precursor	Sigma	10 g	$25
poly-TPD (poly[N,N'-bis(4-butylphenyl)-N,N'-bis(phenyl)benzidine]) — non-polar HTL	Sigma 902388 / Ossila	1 g	$60
(Optional) **TFB** or **PVK** alternative non-polar HTL/host	Ossila	1 g	$50
850 nm long-pass optical filter (photographic / glass)	midopt / Edmund / Amazon	1	$15–$40
Anisole / toluene / hexane (CIS-QD solvents)	—	1 L each	$40

3.2 Removed from hBN BOM (not needed)

- hBN powder, NMP, PMMA-as-hBN-passivation. **Carbon CQD reagents (citric acid, urea) are NOT used** for the emitter (replaced by CIS QD reagents above) — though you may still keep a small carbon-CQD batch as a green reference if desired.

3.3 Equipment deltas

ITEM	FUNCTION	APPROX. COST
Schlenk line / vacuum-Ar manifold (or Ar glovebag + needle-valve flask)	inert-atmosphere hot-injection for CIS QDs	$150–$300 (used Schlenk line) / $40 glovebag
3-neck round-bottom flask + heating mantle + thermocouple (to 300 °C)	hot-injection reactor	$60
Tube furnace (to 1200 °C)	V_Si activation anneal (800–1200 °C) — *needed unless you buy V_Si-ready wafer*	$400–$700
Glovebox (optional)	oxygen/moisture-free CIS handling	(optional, $0 if using glovebag)

Everything else (hotplate, autoclave, sonicator, spin coater, UV, Peltier, sourcemeter, lock-in, ADF4351) is unchanged. The 650 nm filter is **replaced** by the 850 nm long-pass.

Section 1 — CIS/ZnS Quantum-Dot Emitter Array (CHANGED: red/NIR, not green)

Goal: Synthesize **$CuInS_2$/ZnS (CIS) colloidal quantum dots** emitting ~730–760 nm and integrate them into a patterned ITO/PEDOT:PSS/poly-TPD/CIS-ZnS/ZnO/Ag electroluminescent QLED array (8 × 8 pixels, 1 mm² each) on a 25 × 25 mm² ITO-glass substrate. The red/NIR emitter doubles as the on-chip pump for V_Si (no external 730 nm laser). Target: EL peak 730–760 nm, turn-on 3–8 V, detectable ASE/edge narrowing under overdrive.

> **Note.** CIS QLEDs are an established NIR-QLED family (Cd/Pb-free). True electrically-pumped *lasing* from CIS QDs is **not claimed** — as with the carbon CQD chip, the target is **electroluminescence and ASE / high-brightness edge emission**, used as a compact pump. This synthesis is **more demanding than the carbon-CQD autoclave route** (it needs an inert atmosphere and hot injection), but it is the only way to reach 730–760 nm on-chip without a discrete diode laser.

1.1 Hot-injection synthesis of $CuInS_2$/ZnS (CIS) QDs

This is the standard **$CuInS_2$ core + ZnS shell** colloidal route, size-tuned to ~730–760 nm (larger cores → redder emission).

1. **Charge (under Ar).** In a 3-neck flask on the Schlenk line (or Ar glovebag), load **0.5 mmol CuI (95 mg) + 1.0 mmol $In(OAc)_3$ (292 mg)** in **5 mL oleylamine + 5 mL 1-octadecene** + **1 mL oleic acid**. Degas at **110 °C, 30 min** under vacuum, then back-fill with **Ar**.
2. **Heat to growth temp.** Ramp to **230 °C** under Ar with stirring.
3. **Inject sulfur.** Swiftly inject **1 mL 1-dodecanethiol** (or 2 mL of 0.1 M S in oleylamine). The solution darkens (nucleation). Hold at **230 °C, 10–30 min — longer time / higher T → larger cores → redder emission**. For ~730–760 nm, target ~20 min at 230–240 °C.
4. **ZnS shelling (passivation).** Cool to ~200 °C. Inject **2 mL of 0.2 M zinc stearate in ODE** (or zinc acetate + sulfur in oleylamine). Hold **200 °C, 20 min** to grow a ~1–2 monolayer ZnS shell. The shell raises PL quantum yield to **30–60 %** and shifts emission slightly red.
5. **Cool + purify.** Cool to < 60 °C. Precipitate with **ethanol/acetone** (3× volume), centrifuge **5000 rpm, 5 min**, discard supernatant. Redisperse pellet in **hexane or toluene** (~10 mg/mL). Repeat precipitation/redispersion 2× to remove free ligand.
6. **Size/colour check.** Dilute 1:100 in hexane; under 365 nm UV expect **deep-red/NIR emission** (~700–780 nm). Approximate QY vs a standard. For 730–760 nm, cores are ~3.5–4.5 nm; adjust growth time in step 3 to tune.
7. **Wavelength tuning.** CIS emission is tuned by **core size** (growth T/time) and the Cu:In ratio: lower Cu:In (e.g. 1:4) blue-shifts toward 600 nm; stoichiometric 1:2 with larger cores + ZnS reaches ~730–800 nm. **Target ~730–760 nm** so the 850 nm long-pass filter cleanly separates pump from the 916 nm PL.

1.2 CIS-QD electroluminescent ink (non-polar host)

8. **Composite.** Dissolve **poly-TPD** (the non-polar hole-transport layer, also acts as a soft host) in **chlorobenzene or toluene** at **8 mg/mL**. Add **CIS-ZnS QDs** to ~**10–15 mg/mL** final QD concentration, sonicate 5 min, filter 0.45 µm PTFE. (Unlike the carbon-CQD/PVA aqueous ink, CIS QDs are hydrophobic/oleylamine-capped and **do not disperse in water/PVA** — use a non-polar aromatic solvent + poly-TPD host. A ligand-exchange to a short thiol can make CIS QDs polar-dispersible for a PVA host, but it usually quenches PL; the non-polar QLED stack is the higher-yield route.)

1.3 Substrate preparation (ITO glass)

9. Cut ITO glass to 25 × 25 mm^2, pattern the ITO anode by tape/photoresist + $HCl:HNO_3:H_2O$ (1:1:5) etch, 50 °C, 5 min. Clean: sonicate 10 min each in detergent, DI, acetone, IPA. UV-ozone 15 min.

1.4 PEDOT:PSS + poly-TPD hole-injection/transport layers

10. Filter PEDOT:PSS (AI 4083) 0.45 µm, spin on ITO **3000 rpm, 30 s** → ~40 nm. Bake **120 °C, 15 min**.
11. **poly-TPD layer (non-polar bridge).** Spin poly-TPD in chlorobenzene (8 mg/mL) at **2000 rpm, 40 s** → ~30–40 nm. Bake **150 °C, 30 min**. This bridges the polar PEDOT:PSS to the non-polar CIS-QD layer and is **standard in red/NIR QLEDs**; it replaces the direct-CQD-on-PEDOT layout of the carbon chip.

1.5 CIS-ZnS emissive layer

12. Spin the CIS-QD/poly-TPD ink at **2000 rpm, 40 s** → ~60–80 nm. Bake **80 °C, 20 min** under N_2 / vacuum desiccator (oxygen and moisture quench CIS PL).
13. **ASE edge strip (optional):** drop-cast a thicker (~250 nm) neat-CIS strip at the pixel edge through a 0.5 mm shadow mask, dry 60 °C, to form the in-plane gain strip for edge ASE.

1.5b ZnO electron-transport layer (ETL) — unchanged chemistry

14b. ZnO nanopowder in ethanol/butanol 20 mg/mL, sonicate 30 min, 0.45 µm. Spin on the CIS layer **3000 rpm, 30 s** → ~30–50 nm. Bake **150 °C, 10 min**. ZnO (bandgap 3.37 eV ≈ 368 nm) is **transparent to the 730–760 nm pump and the 916 nm PL**; it injects electrons and blocks holes, completing the p-i-n stack.

1.6 Top cathode — unchanged

15. **Top cathode (Ag).** Maker route: dispense **conductive silver paste from a syringe** through the 8 × 8 shadow mask, doctor-blade ~1–2 µm, cure 60–80 °C, 15 min. Lab route: thermal-evaporate Ag 100 nm through mask. Pattern 8 × 8 pads.
16. Encapsulate: spin 200 nm PMMA, cure 120 °C, 10 min.

1.7 Electrical drive and EL check

17. Contact ITO anode + Ag cathode; drive DC, increase 0 V. Turn-on **3–8 V**; red/NIR EL visible under dim light / with a night-vision viewer or phone camera (730–760 nm is at the edge of Si-camera sensitivity). Drive 2–20 mA/pixel.
18. **ASE probe:** drive a pixel at 2× threshold; collect edge emission via fiber into a USB spectrometer (NIR-capable, or a Si spectrometer with reduced sensitivity — note 916 nm is near the Si detector cutoff; an InGaAs spectrometer is ideal for the PL). Look for FWHM narrowing (from ~100 nm to < 30 nm) above threshold = ASE.

Timing, Section 1: CIS synthesis ~1 day (hot-injection + shell + purify) + film stack 1 day + cathode half day. **Total: ≈ 2.5–3 working days** (comparable to the carbon-CQD route but with the Schlenk step).

Section 2 — 4H-SiC Wafer + V_Si Defect Engineering (REPLACES the hBN section)

Goal: Obtain a 4H-SiC wafer sample hosting V_Si (V2) centres ($2|D| \approx 70$ MHz, ZPL 916 nm) at ensemble density for 8 × 8 site addressing, with the surface ready for the emitter/waveguide/detector stack.

2.1 Procurement

1. **Buy the wafer.** Order a **4H-SiC** sample (semi-insulating or N-doped, 5×5 mm to 2-inch, double-side polished). 4H-SiC power-electronic wafers are mass-produced and inexpensive. Expect ~$30–$120.
2. **Native V_Si.** V_Si (V2) are **often present as-grown** in commercial 4H-SiC at low density, and are reliably created/enriched by irradiation. Check vendor data or plan an irradiation step.

2.2 Defect engineering (V_Si creation)

Route A — Irradiation + anneal (recommended, the established route).

3. **Irradiate.** Electron (2 MeV, ~10^{18} cm^{-2}) or proton irradiation creates silicon vacancies. University irradiation services are often free/cheap; commercial services ~$200.
4. **Activation anneal.** Anneal in **Ar or N_2** at **800–1200 °C, 1–2 h** in the tube furnace to mobilize/activate the vacancies and remove damage. *This needs the tube furnace — the hotplate cannot reach it.* SiC is robust up to ~1600 °C so the anneal window is wide and forgiving (much easier than the 800–900 °C narrow hBN window).

Route B — As-grown (no furnace, if the wafer already shows V_Si). Some 4H-SiC wafers show V_Si ODMR as-delivered; if you confirm a PL/ODMR signal, skip irradiation/anneal and proceed to clean.

2.3 Surface cleaning

5. **Solvent clean.** Sonicate 5 min each in acetone, IPA, DI water.
6. **Acid/oxide clean.** Dip in **49 % HF : DI = 1:10** for 60 s (removes native SiO_2; **HF — calcium gluconate gel on hand, full PPE**), or RCA-1 (NH_3:H_2O_2:H_2O) if you want a hydrophilic oxide. Alternatively **O_2 plasma** 100 W, 10 min (dry). Rinse DI, N_2 dry. SiC surfaces are robust and easy to prepare (no boiling triacid needed, unlike diamond).

2.4 Site density and surface proximity

7. **Density.** Irradiated/enriched V_Si ensemble density ~10^{10}–10^{12} cm^{-3} is typical; at the wafer surface this gives resolvable ensemble spots for 8 × 8 ODMR.
8. **Depth caveat.** V_Si in bulk SiC are **distributed through the ~0.5 mm wafer** (even deeper/denser than diamond, since SiC irradiation defects form throughout). Surface proximity for sensing is the **worst** of the three platforms — bulk V_Si are far from the surface. (Shallow V_Si via low-energy implant is possible but outside maker scope.) For the QSSC compute/sensor demonstrator this is acceptable; the trade-off vs hBN is **worst surface proximity, ~20× better coherence, easiest host fabrication**.

2.5 No PMMA-on-host step

9. No PMMA passivation on the SiC. Proceed directly to waveguide/emitter/detector deposition (Section 4). Planarize with SU-8 if the polished surface is too rough for waveguide adhesion.

Timing, Section 2: clean 1 h + (optional) irradiation (outsourced) + anneal 4 h. **Total: ~0.5–1 working day + irradiation turnaround** (the easiest host in the QSSC family).

Section 3 — On-Chip Fluorescence Photodetector Array (Graphene / MoS_2) (CHANGED: graphene-led for IR)

Goal: N-electrode graphene/MoS_2 photodetector array on-chip, FPGA-read, software pulse-capture calibration (Section 5.6). **Identical process to the hBN protocol Section 3**, with one platform note: the 916 nm V_Si PL is in the IR where **monolayer MoS_2 (bandgap ~690 nm) does not absorb**, so the MoS_2 photogating gain is lost. Options: - **Graphene-led channel** (broadband, absorbs 916 nm) — the simplest; bare-graphene responsivity ~0.01–0.1 A/W, detectable with the lock-in. - **Multilayer MoS_2** (few-layer, bandgap ~1.2 eV ≈ 1030 nm) — absorbs 916 nm, restores some photogating gain; use few-layer MoS_2 instead of monolayer for this variant. - **Graphene/MoS_2-photogating heterostructure with few-layer MoS_2** — best of both: graphene collects, few-layer MoS_2 gates.

- CVD graphene transfer, Au/Cr source-drain electrodes, O_2-plasma isolation, bias 0.1–1 V, lock-in readout — as in the hBN base protocol. No discrete photodiodes, no external camera. **Use few-layer MoS_2 (not monolayer) for this variant.**

Section 4 — Waveguides, Stacking (bulk geometry), Filter, Thermal, Contacts

4.1 SU-8 waveguides (on SiC)

1. SiC index n ≈ 2.6 (very high); spin a ~5 µm SU-8 planarization layer (1000 rpm, flood-expose, hard-bake) so waveguide modes are not pulled into the wafer. Then spin SU-8 2002 core (2000 rpm → ~2 µm), pattern 5–10 µm wide guides, PDMS cladding (n 1.41) over SU-8 (n 1.58). Same as the hBN protocol modulo the thicker planarization.

4.2 Coupling the CIS emitter to the waveguide + 850 nm long-pass filter

2. Cleave the CIS stack so the red/NIR gain-strip edge faces the SU-8 input facet; butt-couple in the jig; UV-cure NOA 61. Coupling 20–60 %.
3. **850 nm long-pass filter.** Place an **850 nm long-pass** in the collection path (between the SiC PL output and the detector array, or as a coating on the detector facet). It **passes the 916 nm V_Si PL and blocks the 730–760 nm CIS pump**. (The hBN chip's 650 nm filter is **not** used here — it would pass the 730–760 pump.) If pumping at the red edge (~760 nm), verify the 850 nm filter edge is steep enough; pumping at ~730 nm gives the cleanest separation.

4.3 Stacking (bulk, not 2D)

4. **Stack order (bottom→top):** Si/SiO_2 carrier → Peltier TEC → **4H-SiC wafer** (face up) → SU-8 waveguides → PDMS → CIS emitter chip (edge-coupled) → 850 nm LP filter → graphene/few-layer-MoS_2 detector at waveguide output. The SiC replaces the hBN chip; the rest mirrors the hBN stack. Bond corners with PDMS/cyanoacrylate; cure 80 °C, 1 h. Wire-bond/silver-paste to PCB fan-out.

4.4 Peltier thermal management (unchanged)

5. 2-stage TEC (or TEC1-12706), NTC, PID **295 K ± 0.1 K**, < 5 W. SiC thermal conductivity (~370 W/m·K) is excellent; temperature stability is easy.

4.5 Contacts and encapsulation (unchanged)

6. Au wire / silver-paste to header: ITO anode (×8 rows), Ag cathode (×8 cols), detector drains (×N), gate (×1), thermistor (×2), TEC ± (×2). PDMS encapsulation, output facet exposed.

4.6 Coplanar waveguide (MW) for ODMR (unchanged, frequency changes)

7. Hand-etch 50 Ω CPW on FR4/Rogers; mount the SiC wafer face-down over the signal line (≤ 50 µm gap) or flip-chip with anisotropic paste. **Microwave frequency changes to 70 MHz** (V_Si V2, 2|D| = 70 MHz) — well within the ADF4351 range (35 MHz–4.4 GHz); at 70 MHz you may alternatively use a low-frequency function generator directly. **No hardware change** to the PLL chain.

> **Optional divacancy upgrade (not the primary protocol):** the SiC **divacancy** V_SiV_C (S=1, D ≈ 1.3 GHz, ZPL ~1.1 µm, T_2 ~ ms with DD) would compute small Shor at room temperature. It needs a **~808/980 nm pump** (CIS QDs reach ~800; for 980 use PbS — Pb, toxic — or a 980 diode) and a **~1000 nm long-pass filter**. The QSSC simulator profiles V_Si (70 MHz), so this variant stays V_Si-primary; the divacancy is documented as a future upgrade path.

Section 5 — Characterization & Testing

5.1 Electroluminescence (CIS-QD array)

1. Drive each pixel 0–12 V, confirm turn-on 3–8 V and red/NIR EL (~730–760 nm; check with a night-vision viewer, phone camera, or InGaAs/Si spectrometer). ASE check: FWHM narrowing < 30 nm above threshold. As in the hBN protocol.

5.2 ODMR of V_Si in 4H-SiC

4. **Optical:** on-chip CIS-QD emitter (~730–760 nm, ~1 mW at the sample) — no external laser. Collect V_Si PL at **~916 nm** through the **850 nm long-pass filter** (passes 916, blocks the 730–760 pump) into the **on-chip graphene/few-layer-MoS_2 detector array**. No Si APD / discrete photodiode. (A Si spectrometer has reduced sensitivity at 916 nm near its cutoff; an InGaAs detector/spectrometer is preferred for characterization, but the on-chip graphene detector reads 916 nm fine.)
5. **Microwave:** sweep **~50–90 MHz** from the function generator / ADF4351 through the CPW. Modulate MW at 1 kHz for lock-in.
6. **Expected resonance:** a resonance at **2|D| ≈ 70 MHz** (V_Si V2, S=3/2; the ^{29}Si hyperfine, A ~ 1.5 MHz, is weak/sub-linewidth and usually merges into one line). ODMR contrast **1–10 %** (ensemble), linewidth **5–30 MHz**. The S=3/2 quartet structure splits under a static B_0.
7. **Magnetic field sweep:** apply B_0; observe the S=3/2 sublevel splitting, D shift $D(B) = D_0 \pm \gamma_e \cdot B_0$, $\gamma_e/2\pi \approx 28$ GHz/T — confirms the spin-3/2 assignment.

5.3 Spin coherence (T_2)

8. **Pulsed ODMR** (Hahn echo at 70 MHz). Vary τ from 0.1 µs to **100 µs**. Fit echo vs τ to exp[−(2τ/T_2)^n].
9. **Expected T_2: 20–60 µs at room temperature** (natural abundance, ensemble), up to **~100 µs isotopically purified** (^{4}H-^{28}SiC), extended with dynamical decoupling. T_1 typically **~340 µs**. This is **~20× longer than V_B$^-$** — enough for shallow algorithms (Bell pairs, small QFT, Ramsey sensing) at 300 K, but **deep algorithms (Shor N=15) do not fit T_2 at 300 K without DD + isotopic purification**; cool to raise T_2 further.

5.4 Coupling efficiency (waveguide → detector) — unchanged

Inject calibrated 730 nm light into the SU-8 facet; measure output + on-chip detector photocurrent through the 850 nm LP. Target 20–60 % coupling, < 1 dB/cm loss.

5.5 End-to-end functional test

12. Drive the CIS pixel electrically → red/NIR EL pumps the V_Si ensemble → collect 916 nm PL (through the 850 nm long-pass filter) at the on-chip graphene/few-layer-MoS_2 detector → photocurrent modulated by MW at **70 MHz** → FPGA/lock-in reads the ODMR dip. Successful end-to-end ODMR with the **on-chip CIS-QD pump** (no external laser, no big diode) is the chip's defining demonstration.

5.6 Emitter ↔ qubit ↔ detector calibration (software pulse-capture) — unchanged

13. Same automated FPGA sweep: for each emitter pixel fire a red/NIR pump pulse, sample every detector electrode, record which sees a modulated 916 nm response under MW at **70 MHz** → build the emitter-pixel → qubit-site → detector-electrode lookup table. No camera, no alignment jig. (The MW frequency in the sweep is 70 MHz, the filter is 850 nm LP, the pump is red/NIR — the only calibration deltas from the hBN chip.)

Section 6 — Safety

HAZARD	CONTROL
Schlenk line / hot flask 230–240 °C, Ar	Heat-resistant gloves; confirm Ar flow; vent overpressure; fire risk with organics
Oleylamine / 1-dodecanethiol / ODE (high-bp organics, odorous)	Fume hood; no ignition; capture waste
HF (49 %, dilute) for SiC clean	**Calcium gluconate gel on hand**, face shield, butyl gloves, fume hood
HNO_3, HCl	Fume hood, PPE
Tube furnace 800–1200 °C	Heat-resistant gloves; cool before opening; fire risk
Electron/proton irradiation (outsourced)	Facility controls; treat irradiated wafer per facility guidance
730–760 nm high-brightness EL	Not class 3B (it's an LED), but avoid staring into the bright NIR source; NIR is invisible — use a viewer
MW up to 4 GHz, < +20 dBm	Low risk; keep > 30 cm
Peltier / hot surfaces	Heat-resistant gloves
Nanomaterials (CIS QDs, In/Cu salts, graphene/MoS_2)	Indium/copper salts are moderately toxic — N95, gloves, fume hood for synthesis; dispose CIS waste as heavy-metal solid waste

7. Expected Performance Table (SiC V_Si variant)

PARAMETER	TARGET / EXPECTED	BASIS
CIS-QD pump (on-chip)	**730–760 nm** (red/NIR)	Sec. 1
CIS-QD EL turn-on	3–8 V	Sec. 1
CIS-QD lasing	**NOT claimed** (ASE only)	see note
Host	4H-SiC wafer (bulk 3D)	Sec. 2
Defect	V_Si (V2), S=3/2, 2	D
V_Si PL	~916 nm (NIR)	Sec. 5.2
Optical filter	**850 nm long-pass (CHANGED)**	Sec. 5.2
Defect density	~10^{10}–10^{12} cm^{-3} (ensemble)	Sec. 2.4
ODMR contrast	1–10 %	Sec. 5.2
ODMR linewidth	5–30 MHz	Sec. 5.2
Spin coherence T_2 (RT)	**20–60 µs (≤100 µs isotopic)**	Sec. 5.3
Spin relaxation T_1 (RT)	~340 µs	Sec. 5.3
Compute at 300 K?	Shallow only (Bell, small QFT, Ramsey); deep (Shor) needs DD + cool	Sec. 5.3
Waveguide coupling	20–60 %	Sec. 4.2
Graphene detector responsivity	0.01–0.1 A/W (few-layer MoS_2 restores some gain)	Sec. 3
Operating temperature	288–300 K (set 295 K)	Sec. 4.4
TEC power	< 5 W	Sec. 4.4
Chip footprint	25 × 25 mm^2 (host = 4H-SiC wafer ~5×5×0.5 mm)	Sec. 2

8. Validation Against Known Studies

8.1 V_Si (V2) in 4H-SiC — D, ODMR, coherence

- **V_Si (V2), 2|D| = 70 MHz, S=3/2, room-temperature spin control:** Kraus et al., *Nature Physics* 10, 157–162 (2014) — the basis for the 70 MHz figure and the S=3/2 assignment used by the QSSC simulator `SiC-VSi` profile.
- **Coherent control of single spins in SiC at room temperature:** Widmann et al., *Nature Materials* 14, 164–168 (2015).
- **V_Si as a quantum system / single-defect spectroscopy:** Baranov et al., *PRB* 83, 125203 (2011).
- **Single V1 spin-optical dynamics:** Morioka et al., *Phys. Rev. Applied* 17, 054005 (2022).
- **Room-T ensemble spin echo of V_Si, field-dependent T_2 (tens of µs):** Carter et al., *PRB* 92, 161202 (2015).
- **V_Si T_1 ~ 340 µs, T_2 ~ 104 µs at 295 K (proton-irradiated 4H-SiC):** Embley et al., *PRB* 95, 045206 (2017) — the basis for T_2 ≈ 40 µs (natural) / ~100 µs (purified).
- **Isotopic purification extends V_Si T_2 to ~105 µs:** Lekavicius et al., *PRX Quantum* 3, 010343 (2022).
- **Divacancy (V_SiV_C, S=1, D ≈ 1.3 GHz, ms coherence) — distinct defect, optional upgrade:** Christle et al., *Nat. Mater.* 14, 160–163 (2015) and *PRX* 7, 021046 (2017).
- **Dynamical decoupling to extend T_2:** de Lange et al., *Science* 330, 60–63 (2010); Ryan et al., *PRL* 105, 200402 (2010).

8.2 CIS/ZnS QDs and NIR QLEDs

- **$CuInS_2$/ZnS core/shell colloidal QDs, size-tunable 550–800+ nm, Cd/Pb-free:** the CIS/ZnS QD literature (e.g. Zhong et al. and subsequent CIS-QD reviews) establishes the synthesis and the 550–800 nm tunability used in Section 1.
- **CIS-QD electroluminescent NIR QLEDs (ITO/PEDOT:PSS/poly-TPD/CIS/ZnO/metal):** the poly-TPD interlayer + ZnO ETL stack is the standard red/NIR QLED architecture, supporting Section 1.4–1.5b.
- CIS QDs are chosen specifically because **carbon quantum dots (citric-acid+urea) do not reach 730–760 nm** (their ceiling is green ~540 nm).

8.3 Detector / waveguides / Peltier

- Graphene (broadband, reads 916 nm IR) and few-layer MoS_2 (bandgap ~1.2 eV, absorbs ~1030 nm and thus 916 nm) detector performance is consistent with the hBN base protocol validation (graphene: Xia et al. 2009; MoS_2: Lopez-Sanchez et al. 2013; photogating: Konstantatos et al. 2012). SU-8/PDMS waveguide coupling and Peltier stability are as in the hBN base.

End of SiC V_Si variant. This document is a maker-lab demonstrator specification. It reuses the hBN chip's detector concept, waveguides, FPGA and software calibration; it changes the on-chip emitter (green carbon CQD → red/NIR CIS/ZnS QDs, on-chip, no big diode), the optical filter (650 → 850 nm long-pass), the host (hBN film → 4H-SiC wafer) and the MW frequency (3.46 GHz → 70 MHz). Where primary literature is cited, consult the cited paper before relying on a specific numerical parameter.

References (platform-specific; CQD/detector/waveguide refs are in the hBN base)

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- H. Zhong et al., and the $CuInS_2$/ZnS colloidal-QD / NIR-QLED literature (size-tunable 550–800 nm, Cd/Pb-free).
